

AI Model Selection Decision Matrix

Balancing Size, Accuracy, Speed, and Explainability Across AI Domains

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Artifact Purpose

A professional portfolio presentation that compares pre-trained models across NLP/Generative AI, Computer Vision, and Tabular Data using a decision matrix for practical model selection.

Purpose of This Artifact

This presentation organizes popular AI models into domains and compares their trade-offs using a clear decision-matrix format.

Portfolio Artifact Goal

This artifact helps students and early AI practitioners choose models based on the problem type, not just model popularity.

Classifies models across NLP/Generative AI, Computer Vision, and Tabular Data.

Compares trade-offs between size, accuracy, inference speed, and explainability.

Shows why the most accurate model is not always the best model for deployment.

Connects model choice to practical use cases and business constraints.

Why Model Selection Matters

Modern AI teams rarely select models based only on accuracy. A model must also fit the domain, infrastructure, speed requirements, and explainability needs of the use case.

Simple Definition

Model selection is the process of choosing the AI model that best balances performance, cost, speed, and trust for a specific problem.

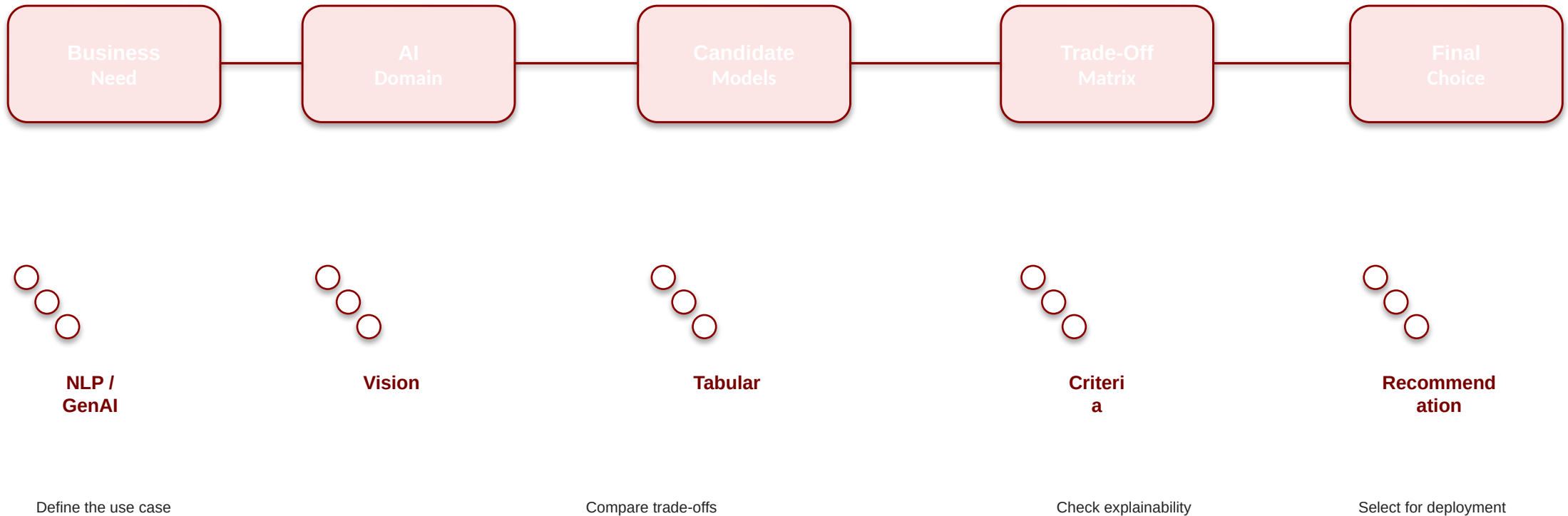
Common model-selection tasks:

- Text generation
- Text classification
- Image classification
- Mobile vision
- Business risk prediction

Main trade-off drivers:

- Model size and memory
- Benchmark accuracy
- Inference speed
- Deployment cost
- Explainability and governance

Overall AI Model Selection Framework



Component 1: Evaluation Criteria for Model Selection

Pre-trained models can be compared using four practical criteria: model size, accuracy, inference speed, and explainability. These criteria affect cost, usability, deployment, and stakeholder trust.

Example

GPT-4 may provide stronger generation capability, but BERT may be faster and easier to deploy for enterprise text classification.

Key Point

The best model is not always the largest model. The best model is the one that fits the domain, constraints, and decision risk.

Component 2: Size, Accuracy, Speed, and Explainability

Each criterion answers a different deployment question. Together, they create a practical model-selection framework.

use case → domain → candidate models → trade-off scores → recommendation

Size affects memory, energy, and deployment cost.

Accuracy affects prediction quality and business value.

Speed affects user experience and real-time feasibility.

Explainability affects trust, governance, and adoption.

Component 3: Model Domains and Typical Use Cases

Definition

Different AI domains require different model families. NLP models process language, vision models process images, and tabular models work best on structured rows-and-columns data.

NLP / Generative AI: GPT-4 for content generation; BERT for text classification.

Computer Vision: EfficientNet for high-accuracy image tasks; MobileNet for mobile and edge devices.

Tabular Data: XGBoost and LightGBM for structured business, risk, and analytics problems.

This classification reveals that model choice should start with the data type and business goal.

A model that performs well in one domain may be a poor fit in another domain. Model selection begins with matching the model family to the data type.

Component 4: Pre-Trained Model Decision Matrix

Model	Domain	Size	Accuracy	Speed	Explainability	Best Fit
GPT-4	NLP / GenAI	Very Large	Very High	Moderate	Low	Complex generation + reasoning
BERT-base	NLP	Medium (110M)	High	Fast	Moderate	Text classification + search
EfficientNet-B0	Vision	Medium (5.3M)	Very High	Moderate	Moderate	High-accuracy image tasks
MobileNetV2	Vision	Small (3.4M)	High	Very Fast	Moderate	Mobile / edge vision
XGBoost	Tabular	Small / flexible	High	Fast	High	Structured risk prediction
LightGBM	Tabular	Small / flexible	High	Very Fast	Medium	Large-scale tabular data

Main Pattern

Large models usually improve capability but reduce speed and explainability. Smaller tabular and vision models are often better for fast, explainable deployment.

Component 5: Explainability and Model Trust

Definition

Explainability describes how easily humans can understand why a model produced a result. It matters most when decisions affect safety, money, health, or compliance.

LLMs are powerful but harder to interpret because decisions are distributed across billions of parameters.

Tree-based models are easier to explain because decisions follow splits, feature importance, and structured rules.

Vision models sit in the middle: they can be inspected using attention maps or saliency methods, but explanations are still approximate.

Model Type	Explainability Pattern
LLMs	Low: opaque internal reasoning
Vision	Medium: visual attribution tools
Tree-based	High: feature splits and importance

Key Idea

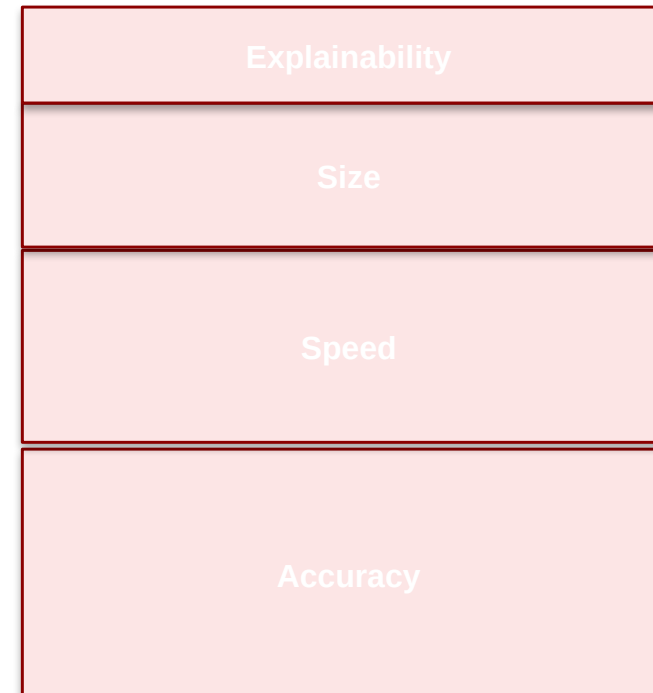
Explainability is a model-selection factor, not an afterthought.

Component 6: Deployment Trade-Offs

Why Trade-Offs Matter

A model that performs well in testing may still be a poor production choice if it is too slow, expensive, or difficult to explain.

GPT-4: strong capability, high cost, low transparency.
BERT: efficient NLP baseline for classification tasks.
MobileNet: fast vision model for mobile and edge devices.
XGBoost: strong balance for structured business data.
LightGBM: speed-focused option for large tabular datasets.



Real-world model selection balances technical performance with operational constraints.

How Model Selection Works



Model selection is an iterative decision process. Teams choose a model, evaluate trade-offs, test it against business needs, and monitor performance after deployment.

Selection Cycle

Define the problem, compare candidate models, validate performance, and revisit the choice as data and deployment needs change.

Model Deep Dive: GPT-4 and BERT

GPT-4 is best suited for complex language generation, reasoning, summarization, coding assistance, and multimodal workflows. BERT is better suited for focused NLP tasks such as classification, search, and sentiment analysis.

Model Snapshot

GPT-4: very large, high capability, moderate speed, low explainability

BERT-base: 110M parameters, high NLP accuracy, faster inference, moderate explainability

Key Point

For enterprise NLP, BERT may be more practical than GPT-4 when the goal is classification rather than open-ended generation.

Model Deep Dive: Vision and Tabular Models

EfficientNet and MobileNet support image tasks, while XGBoost and LightGBM support structured business data. Each family makes a different trade-off between accuracy, speed, size, and explainability.

Model Snapshot

EfficientNet-B0: 5.3M parameters, strong image accuracy
MobileNetV2: 3.4M parameters, optimized for mobile speed
XGBoost: accurate and explainable for structured data
LightGBM: fast and scalable for large tabular datasets

Key Point

Tree-based models remain strong for structured data because they combine practical accuracy with interpretability and deployment efficiency.

Recommendations by Use Case

Use Case	Recommended Model	Reason
Open-ended content generation	GPT-4	Best capability for flexible language tasks
NLP classification/search	BERT	Strong accuracy with faster inference
High-accuracy image classification	EfficientNet	Better accuracy-size balance for vision
Mobile or edge vision	MobileNetV2	Small and fast for constrained devices
Structured business analytics	XGBoost	Strong accuracy and explainability
Large-scale tabular pipelines	LightGBM	Very fast training and inference

Main Takeaway

Main Takeaway

Model selection should start with the use case. The strongest model is the one that best fits the business need, not simply the one with the highest

Portfolio Artifact Value Proposition

Artifact-Specific Value Proposition

This artifact helps AI students, instructors, and early technology professionals understand how to select models across different AI domains. It demonstrates my ability to classify models, compare trade-offs, and connect technical performance to deployment decisions.

Audience: AIML students, instructors, and beginner AI learners.

Value: Clear model-selection framework across NLP, Vision, and Tabular Data.

Skill demonstrated: Technical communication, decision analysis, model evaluation, and visual classification design.

Reflection: What surprised me was that the most accurate model is not always the best model. Comparing models across domains forced me to think like a data scientist: define the problem first, then choose the model that best balances accuracy, speed, explainability, and cost.

Sources Reviewed

OpenAI GPT-4 Technical Report; Devlin et al. (2019) BERT; Tan & Le (2019) EfficientNet; Sandler et al. (2018) MobileNetV2; Chen & Guestrin (2016) XGBoost; LightGBM documentation and course resources.